

ROOSEVELT ROADS, PR, USA

WMO: 785350

Lat: 18.255N

Lon: 65.641W

Elev: 10

StdP: 101.20

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 11630

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.2	21.2	13.9	9.9	24.2	15.7	11.2	24.8	8.9	26.0	8.3	26.1	1.0	330	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	32.3	26.4	31.9	26.1	31.3	25.8	27.4	30.5	27.1	30.2	26.8	29.9	5.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.1	29.0	26.2	21.7	28.8	26.0	21.3	28.8	86.5	30.7	85.1	30.4	83.8	29.9	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.2	7.5	7.0	DB	17.8	32.9	1.0	0.7	17.0	33.4	16.5	33.8	15.9	34.1	15.2	34.6
			WB	16.1	28.4	1.5	0.8	15.1	28.9	14.2	29.4	13.4	29.8	12.3	30.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	25.2	25.3	25.7	26.4	27.3	28.2	28.5	28.7	28.3	27.8	27.0	25.9
	DBStd	1.61	1.08	1.06	1.12	1.12	1.08	0.90	1.02	0.91	0.92	0.96	1.05	1.06
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6218	472	427	486	492	537	546	574	580	548	553	509	493
	CDD18.3	3176	214	194	227	242	279	296	316	322	298	295	259	235
	CDH23.3	31394	1498	1328	1716	2125	2822	3339	3681	3838	3446	3179	2500	1921
	CDH26.7	8921	188	189	298	454	732	1069	1302	1449	1247	1053	638	300
Wind	WSAvg	3.8	3.9	3.9	4.1	4.0	3.9	4.3	4.4	4.1	3.5	3.1	3.3	3.8
Precipitation	PrecAvg	1442	70	57	64	91	167	97	104	136	152	191	161	120
	PrecMax	2417	212	171	151	267	690	244	215	298	413	696	494	408
	PrecMin	858	22	24	11	12	12	19	38	45	50	52	33	35
	PrecStd	431	38	31	37	58	146	56	48	58	84	125	101	84
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	30.0	30.3	31.2	31.7	32.1	32.7	32.8	32.5	32.4	31.3	30.2
		MCWB	23.0	22.8	24.1	24.2	25.0	25.6	26.7	26.8	26.7	26.3	25.3	24.7
	2%	DB	28.8	29.0	29.3	30.2	31.0	31.3	32.1	32.2	32.0	31.7	30.9	29.2
		MCWB	23.1	22.9	23.4	23.7	24.8	25.5	26.2	26.5	26.5	25.9	25.2	23.9
	5%	DB	28.0	28.4	28.9	29.6	30.2	31.0	31.3	31.5	31.3	31.2	30.1	28.9
		MCWB	23.1	23.0	23.2	23.5	24.6	25.3	25.7	26.2	26.2	25.7	25.0	23.8
	10%	DB	27.7	27.8	28.1	28.9	29.7	30.3	30.9	31.1	31.0	30.6	29.5	28.3
		MCWB	23.0	22.8	22.8	23.5	24.6	25.2	25.5	26.0	26.1	25.5	24.8	23.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.0	25.2	25.7	26.6	26.6	27.1	27.6	27.9	28.0	27.5	26.8	26.3
		MCDB	27.4	27.4	28.0	28.1	29.5	30.0	31.0	31.3	31.2	30.3	29.3	29.0
	2%	WB	24.4	24.3	24.7	25.5	25.9	26.7	26.9	27.3	27.4	27.0	26.2	25.3
		MCDB	27.2	27.4	27.9	28.2	28.9	29.7	30.4	30.6	30.4	29.9	28.9	27.9
	5%	WB	24.0	23.9	24.2	24.7	25.6	26.3	26.6	26.9	27.0	26.7	25.7	24.7
		MCDB	27.0	27.1	27.4	27.9	28.7	29.4	30.1	30.2	30.0	29.6	28.6	27.4
	10%	WB	23.7	23.5	23.8	24.3	25.2	25.8	26.1	26.7	26.7	26.3	25.3	24.4
		MCDB	26.6	26.7	27.0	27.6	28.4	29.0	29.6	29.9	29.7	29.2	28.4	27.2
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.6	5.7	5.6	5.2	4.9	5.0	5.2	5.9	6.0	5.6	5.2
		MCDBR	5.7	6.1	6.1	6.0	5.6	5.3	5.5	5.6	6.3	6.5	6.2	5.5
		MCWBR	2.9	2.9	2.9	2.8	2.6	2.4	2.7	2.8	3.0	3.0	2.9	2.7
	5% WB	MCDBR	5.3	5.7	5.7	5.7	5.2	4.9	5.2	5.3	5.9	6.1	5.5	5.1
		MCWBR	2.8	3.0	2.9	3.0	2.7	2.6	2.9	2.9	3.1	3.2	3.0	2.7
Clear-Sky Solar Irradiance	taub	0.376	0.390	0.390	0.415	0.458	0.532	0.533	0.493	0.446	0.397	0.390	0.382	
	taud	2.542	2.494	2.489	2.452	2.338	2.092	2.086	2.228	2.394	2.550	2.562	2.558	
	Ebn at Noon	908	912	920	893	844	776	777	814	856	894	888	891	
	Edh at Noon	98	108	111	116	129	163	164	144	121	100	95	94	
All-Sky Solar Radiation	RadAvg	4.67	5.44	6.06	6.36	6.13	6.20	6.29	6.14	5.79	5.11	4.51	4.35	
	RadStd	0.14	0.12	0.23	0.34	0.40	0.37	0.32	0.24	0.37	0.36	0.32	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.28	+0.47	N/A	+0.33	N/A	N/A	N/A	N/A	+113	+112

Nomenclature: See separate page